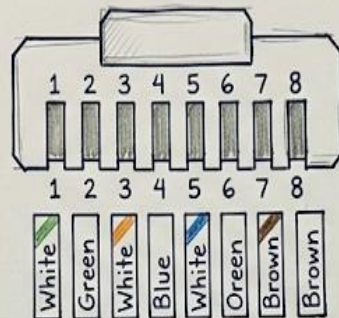


Ethernet Standard and Specification

1. Image of RJ45 connector T568A and T568B Standard :

RJ-45 CONNECTOR PIN LAYOUT (View from bottom, clip UP)
PINS 1-8



T568A Standard	
PIN	WIRE COLOR
1	→ White/Green
2	→ Green
3	→ White/Orange
4	→ Blue
5	→ White/Blue
6	→ Orange
7	→ White/Brown
8	→ Brown

T568B Standard	
PIN	WIRE COLOR
1	→ White/Orange
2	→ Orange
3	→ White/Green
4	→ Blue
5	→ White/Blue
6	→ Green
7	→ White/Brown
8	→ Brown

2. Ethernet Standards Specification Comparison :

Specification	10BASE-T	100BASE-TX (Fast Ethernet)	1000BASE-T (Gigabit Ethernet)
Data Transfer Speed	10 Mbps	100 Mbps	1000 Mbps (1 Gbps)
Minimum Cable Type	Cat3 (or higher)	Cat5 (or higher)	Cat5e / Cat6
Max Cable Length	100 meters	100 meters	100 meters
Connector Type	RJ-45	RJ-45	RJ-45
Twisted Pairs Used	2 Pairs (4 Wires)	2 Pairs (4 Wires)	4 Pairs (8 Wires)

3. Direct Laptop-to-Laptop Ethernet Connection :

- Cable Type Selection

You would traditionally use a **Crossover Cable**.

- **Transmit and Receive Wiring:** In direct device-to-device communication, both laptops are classified as same-tier devices (MDI).
- On a standard **Straight-Through Cable**, Pin 1 connects to Pin 1, and Pin 2 connects to Pin 2. If used directly between two laptops, both would attempt to transmit data over the exact same wires (Pins 1 and 2) and receive on the same wires (Pins 3 and 6), causing a signal collision where neither laptop can hear the other.
- A **Crossover Cable** physically swaps the transmit pairs (TX) on one side to match the receive pairs (RX) on the other side (connecting Pin 1 & 2 on Laptop A to Pin 3 & 6 on Laptop B).

4. Real-World Devices & Supported Ethernet Standards :

- 1. **Modern Gaming Laptop (e.g., ASUS TUF Gaming):**
 - **Supported Standard:** 1000BASE-T (Gigabit Ethernet)
 - **Reason:** High-performance laptops feature gigabit Network Interface Cards (NICs) to handle high-bandwidth needs like massive game downloads and low-latency online gaming.
- 2. **Wi-Fi Router / Optical Network Terminal (ONT):**
 - **Supported Standard:** 1000BASE-T (Gigabit Ethernet)

- **Reason:** Modern routers are equipped with Gigabit LAN/WAN ports so they do not bottleneck high-speed fiber internet connections coming into home or college networks.
- **3. Local Network Printer or POS Terminal:**
 - **Supported Standard:** 100BASE-TX (Fast Ethernet)
 - **Reason:** Network printers and billing terminals transfer very small text/print job files, making 100 Mbps more than enough bandwidth for reliable daily operation.